LOCAL-PROMPT: EXTENSIBLE LOCAL PROMPTS FOR FEW-SHOT OUT-OF-DISTRIBUTION DETECTION

Anonymous Authors

ABSTRACT

Few-shot out-of-distribution detection with vision-language models remains difficult because hard outliers often differ from in-distribution classes only through subtle localized regions, while most prompt-learning methods adapt a single global prompt. We present Local-Prompt, a coarse-to-fine framework that freezes the global prompt branch and learns only local prompts for regional discrimination. The method first generates hard positives and hard negatives from random crops using a frozen global prompt as a selector, avoiding any dependence on real outlier data during training. It then optimizes class-specific local prompts together with a bank of local negative prompts using regional contrastive objectives and a diversity regularizer, and combines global and local evidence through a regional MCM-style score at test time. On ImageNet-1k with four shots, Local-Prompt improves over CoOp from 33.82 to 27.66 FPR95 and from 90.98 to 93.53 AUROC. When plugged into LoCoOp, it further reduces average FPR95 from 31.46 to 26.29 and increases AUROC from 92.68 to 94.03, while also improving in-distribution accuracy. The same local prompting strategy transfers well to harder near-OOD and ImageNet-100 settings, showing that explicitly modeling regional outlier cues is an effective and extensible direction for few-shot OOD detection.

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1 INTRODUCTION

Out-of-distribution (OOD) detection is a core reliability requirement for deploying visual recognition systems in open environments. Vision-language models such as CLIP provide a particularly attractive starting point because they align images and text in a shared embedding space and support open-vocabulary scoring with little or no task-specific training (Radford et al., 2021; Ming et al., 2022). In the few-shot setting, prompt learning further improves adaptation efficiency by updating only a small set of textual context parameters (Zhou et al., 2022b;a). However, few-shot OOD detection remains substantially harder than few-shot classification: many failure cases are near-distribution and differ from in-distribution (ID) classes only in a few localized regions.

This locality matters. A global image-text score can be dominated by the main object category and overlook small but decisive mismatches, such as a background pattern, a local part configuration, or a scene-specific attribute. Existing prompt-learning approaches for few-shot OOD detection, most notably LoCoOp (Miyai et al., 2023a), already exploit local visual features, but the prompt parameters themselves are still optimized at the global level. This creates a mismatch between the granularity of the visual evidence needed for hard OOD detection and the granularity at which prompting is learned.

We address this mismatch with Local-Prompt, a coarse-to-fine prompt-learning framework that separates global and local roles. The global prompt branch is kept frozen and used as a stable selector of informative crops. The trainable capacity is moved into a local branch composed of class-specific local prompts and a bank of local negative prompts. During training, Local-Prompt first produces multiple random crops of each ID image and uses the frozen global branch to identify crops that are highly consistent with the ground-truth class and crops that are least consistent. These serve as hard positives and hard negatives without requiring external OOD examples. The selected crops are then used to optimize regional objectives over top-ranked local regions, encouraging label-consistent local alignment on positive views and explicit rejection on hard negatives.

The design is intentionally extensible. Because local prompting is disentangled from global prompting, the learned local branch can be combined with stronger global prompt learners instead of replacing

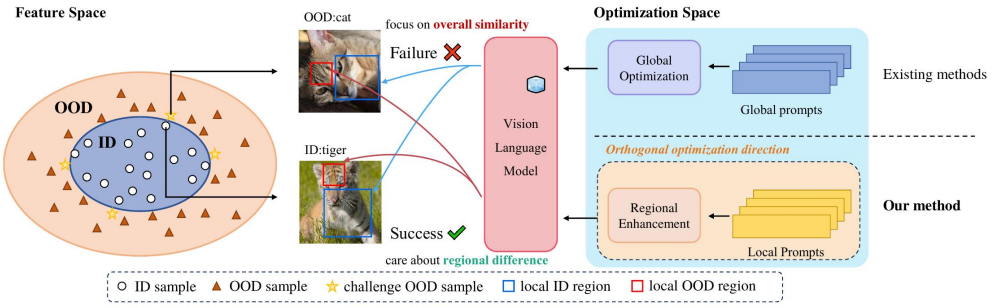

Figure 1: Comparison of prompt learning strategies for OOD detection. Existing few-shot approaches mainly adapt a global prompt and use local visual cues only indirectly. Local-Prompt decouples the two roles: a frozen global branch selects informative crops, while trainable local prompts and local negative prompts learn regional evidence that is more useful for separating subtle OOD samples.

them. This is reflected in the experiments: pairing Local-Prompt with LoCoOp yields consistent gains in both OOD detection and ID accuracy on ImageNet-1k.

Our contributions are threefold:

- We introduce a coarse-to-fine few-shot OOD detection framework that freezes global prompts and learns local prompts specialized for regional outlier cues.
- We propose global-prompt-guided crop selection and a local training objective that combines regional contrastive supervision, hard negative local prompts, and diversity regularization.
- We develop a regional MCM-style inference score and show that the local branch is orthogonal to stronger global prompt learners, yielding additional gains when combined with LoCoOp.

2 Related Work

2.1 Vision-Language OOD detection

CLIP-based OOD detection has shown that aligned image-text representations provide a strong zero-shot signal for open-world recognition (Radford et al., 2021; Ming et al., 2022). MCM scores OODness by the maximum class posterior induced by CLIP concepts (Ming et al., 2022), while follow-up work improves zero-shot robustness through localized scoring or explicit negation semantics (Miyai et al., 2023b; Wang et al., 2023). Other methods expand the negative textual space using textual outlier exposure or large corpora of negative labels (Park et al., 2023; Jiang et al., 2024). These methods are strong, but most are either zero-shot or rely on stronger external negative resources than the few-shot ID-only setting considered here.

2.2 Prompt learning for CLIP adaptation

Prompt learning adapts pretrained vision-language models by optimizing a small number of context vectors rather than fully fine-tuning the encoder (Zhou et al., 2022b;a). This paradigm is especially appealing in few-shot regimes because it preserves most of the prior knowledge in CLIP while remaining data-efficient. Yet standard prompt learning is optimized for classification accuracy, not for separating unseen OOD data. As a result, a globally tuned prompt can still encode nuisance correlations that are harmful for OOD detection.

2.3 Few-shot OOD detection

Among few-shot CLIP-based OOD methods, LoCoOp is most closely related to our work (Miyai et al., 2023a). It regularizes prompt learning with local visual features to suppress ID-irrelevant nuisances in the text embedding. More broadly, few-shot OOD detection has also been studied

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through fully supervised training or synthetic outlier generation, including Energy (Liu et al., 2020), deep nearest neighbors (Sun et al., 2022), and NPOS (Tao et al., 2022). These methods are valuable reference points, but they do not exploit the compositional structure of vision-language prompting in the way Local-Prompt does. Our method differs from LoCoOp and related prompt learners by explicitly allocating the trainable capacity to regional prompts and by constructing hard negatives from the ID images themselves.

3 METHOD

3.1 PROBLEM SETUP

We consider few-shot OOD detection with a pretrained CLIP model (Radford et al., 2021). The training set contains only a few labeled ID examples

$$\mathcal{D}_{\text{ID}} = \{(x_i, y_i)\}_{i=1}^{N},$$

where $y_i \in \{1, \dots, C\}$. At test time, the model must both classify ID examples and reject samples from unseen OOD classes. No real OOD images are used during training.

3.2 GLOBAL AND LOCAL PROMPTS

For each class c, Local-Prompt uses two textual branches. The global branch is frozen and uses the hand-crafted prompt "a photo of a c." In the released implementation, this branch is never optimized and serves as a stable reference. The local branch contains trainable class-specific local prompts $\{p_c^{\ell}\}_{c=1}^{C}$ and a trainable bank of class-agnostic local negative prompts $\{p_j^{-}\}_{j=1}^{N_{\text{neg}}}$.

Given an image x, the CLIP visual encoder returns a global feature $v(x)$ and a set of local region features $\{r_n(x)\}_{n=1}^{N_r}$. The global branch produces logits

$$z_c^{g}(x) = \tau \langle v(x), t_c^{g} \rangle,$$

where t_c^{g} is the frozen global text feature for class c and τ is the learned CLIP logit scale. The local branch produces region-wise logits against class-specific local prompts and negative prompts:

$$z_{n,c}^{\ell}(x) = \tau \langle r_n(x), t_c^{\ell} \rangle, \qquad z_{n,j}^{-}(x) = \tau \langle r_n(x), t_j^{-} \rangle.$$

3.3 STAGE I: GLOBAL-PROMPT-GUIDED CROP SELECTION

The first stage constructs informative positives and negatives from ID images themselves. For each training image, we sample multiple random crops. The frozen global branch scores every crop by the cosine similarity between the crop feature and the ground-truth class text feature. Crops with the highest similarity are treated as positive views because they preserve class-defining evidence, while the least similar crop is treated as a hard negative because the random crop has likely removed or corrupted the most class-relevant region.

This procedure injects OOD-like supervision without requiring external outliers. It is also stable: because the selector is frozen, the selection criterion does not drift together with the trainable local prompts. In practice, the released code uses several highest-scoring crops as positives and the lowest-scoring crop as the negative view.

3.4 STAGE II: LOCAL PROMPT ENHANCED REGIONAL REGULARIZATION

The selected crops are used to optimize local prompts on top-ranked regions. For a positive crop, we first identify the top-k regions according to the logit of the ground-truth class local prompt. The positive local objective encourages those regions to prefer the ground-truth local prompt over all class prompts and over the negative prompt bank:

$$\mathcal{L}_{\text{pos}} = -\log \frac{\sum_{n \in \text{TopK}(y)} \exp(z_{n,y}^{\ell}/T)}{\sum_{n \in \text{TopK}} \left(\sum_{c=1}^{C} \exp(z_{n,c}^{\ell}/T) + \sum_{j=1}^{N_{\text{neg}}} \exp(z_{n,j}^{-}/T) \right)}.$$

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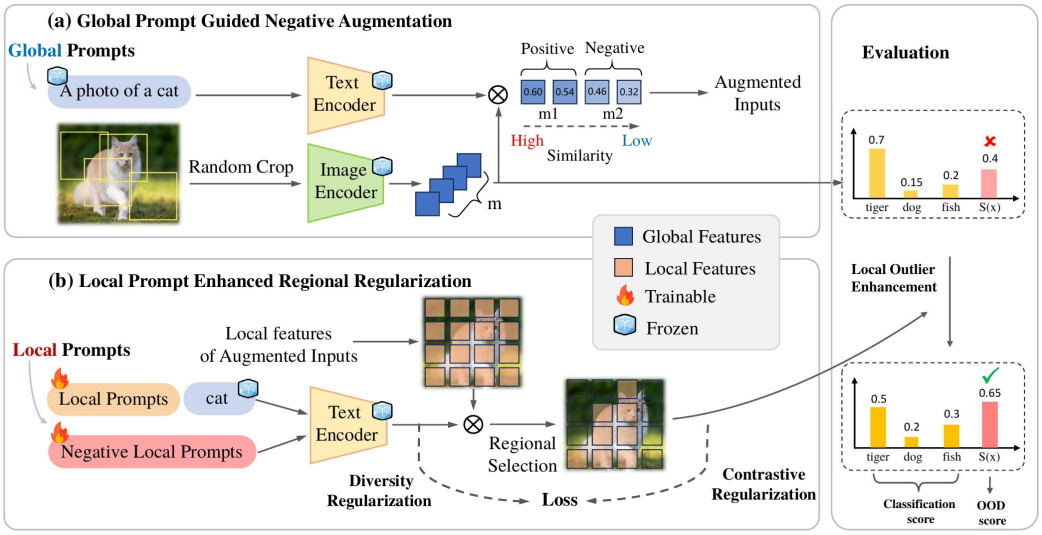

Figure 2: Detailed structure of Local-Prompt. The frozen global branch selects informative crops and provides a stable coarse score, while the local branch learns class-specific regional prompts together with a bank of local negative prompts.

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This term enforces localized evidence for the correct class.

For the hard negative crop, we again focus on the top-k positions indicated by the ground-truth class responses, but now encourage the local negative prompts to dominate those positions:

$$\mathcal{L}_{\text{neg}} = -\log \frac{\sum_{n \in \text{TopK}(y)} \sum_{j=1}^{N_{\text{neg}}} \exp(z_{n,j}^{-}/T)}{\sum_{n \in \text{TopK}} \left(\sum_{c=1}^{C} \exp(z_{n,c}^{\ell}/T) + \sum_{j=1}^{N_{\text{neg}}} \exp(z_{n,j}^{-}/T) \right)}.$$

Intuitively, the model learns that label-inconsistent local evidence should be captured by the negative prompt bank instead of being absorbed into class prompts.

To prevent the negative prompt bank from collapsing, we add a diversity regularizer that penalizes pairwise cosine similarity among negative text features:

$$\mathcal{L}_{\text{div}} = \frac{1}{N_{\text{neg}}(N_{\text{neg}} - 1)} \sum_{i \neq j} \cos(t_i^{-}, t_j^{-}).$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pos}} \lambda \mathcal{L}_{\text{neg}} \beta \mathcal{L}_{\text{div}}.$$

In the released configuration, the local branch is trained for 30 epochs using SGD, with 16 context tokens, class-specific local prompts, and a bank of 300 negative prompts.

3.5 Regional Scoring and Extensibility

At test time, we use different combinations of global and local evidence for classification and OOD detection.

For ID classification, the implementation multiplies the global class score with the average of the top-k regional local scores of the same class:

$$S_c^{\text{ID}}(x) = \exp(z_c^{g}(x)) \cdot \frac{1}{k} \sum_{n \in \text{TopK}_c} \exp(z_{n,c}^{\ell}(x)/T).$$

This keeps the global class semantics while refining them with local evidence.

Table 1: Results when ImageNet-1k is used as ID data. We compare the methods in different tuning manners. † represents the result by our re-implementation. **Bold** values represent the best results.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Zero-shot										
MCM	30.91	94.61	37.59	92.57	44.69	89.77	57.77	86.11	42.74	90.77
GL-MCM	15.18	96.71	30.42	93.09	38.85	89.90	57.93	83.63	35.47	90.83
Full-tuning										
MSP	54.05	87.43	73.37	78.03	72.98	78.03	68.85	79.06	67.31	80.64
Energy	29.75	94.68	53.18	87.33	56.40	85.60	51.35	88.00	47.67	88.90
KNN	29.17	94.52	35.62	92.67	39.61	91.02	64.35	85.67	42.19	90.97
NPOS	16.58	96.19	43.77	90.44	45.27	89.44	46.12	88.80	37.93	91.22
Textual-OE	29.61	94.74	57.12	87.34	66.82	83.71	79.29	77.76	58.21	85.88
Few-shot					4-*shot*					
CoOp	18.95	95.52	29.58	92.90	38.72	89.64	48.03	85.87	33.82	90.98
Local-Prompt	**9.65**	**97.87**	20.40	95.57	29.39	92.67	51.20	88.00	27.66	93.53
LoCoOp†	21.67	95.69	22.98	95.07	31.41	92.10	49.79	87.85	31.46	92.68
Local-Prompt+LoCoOp	12.81	97.29	**19.34**	**95.85**	**27.53**	**92.97**	**45.51**	**89.99**	**26.29**	**94.03**
					16-*shot*					
CoOp	14.60	96.62	28.48	92.65	36.49	89.98	43.13	88.03	30.67	91.82
Local-Prompt	8.71	98.10	23.97	94.85	32.50	92.32	47.93	89.04	28.27	93.58
LoCoOp	16.05	96.86	23.44	95.07	32.87	91.98	42.28	90.19	28.66	93.52
Local-Prompt+LoCoOp	**8.63**	**98.07**	**23.23**	**95.12**	**31.74**	**92.42**	**34.50**	**92.29**	**24.52**	**94.48**

For OOD detection, we start from the MCM score (Ming et al., 2022) on the global branch and add a local rejection term that normalizes local class scores against both local class prompts and local negative prompts:

$$S^{\mathrm{OOD}}(x) = -\max_{c} \mathrm{softmax}(z^{g}(x)/T)_{c} - \frac{1}{k} \sum_{(n,c)\in \mathrm{TopK}} \frac{\exp(z^{\ell}_{n,c}(x)/T)}{\sum_{c'} \exp(z^{\ell}_{n,c'}(x)/T) + \sum_{j} \exp(z^{-}_{n,j}(x)/T)}. \tag{1}$$

We refer to this as a regional MCM-style score. Because the local branch is trained separately from the global branch, it can also be combined with stronger prompt learners. In particular, we evaluate Local-Prompt together with LoCoOp and observe consistent gains.

4 Experiments

4.1 Setup

We follow the large-scale CLIP-based OOD benchmarks centered on ImageNet (Deng et al., 2009). The main far-OOD evaluation uses ImageNet-1k as ID data and iNaturalist, SUN, Places, and Texture as OOD datasets (Van Horn et al., 2018; Xiao et al., 2010; Zhou et al., 2017; Cimpoi et al., 2014). We additionally report ImageNet-100 and near-OOD transfer between ImageNet-10 and ImageNet-20. The implementation is based on CLIP ViT-B/16 (Radford et al., 2021); only the prompt learner is optimized, while the image and text encoders are frozen.

We report FPR95 and AUROC for OOD detection and standard top-1 accuracy for ID classification. The few-shot baselines include CoOp (Zhou et al., 2022b) and LoCoOp (Miyai et al., 2023a). We also compare against representative zero-shot and fully tuned detectors when those results are available in the provided experiment tables, including MCM (Ming et al., 2022), GL-MCM, MSP (Hendrycks & Gimpel, 2016), Energy (Liu et al., 2020), deep nearest neighbors (Sun et al., 2022), NPOS (Tao et al., 2022), and Textual-OE (Park et al., 2023).

4.2 ImageNet-1k Results

Table 1 shows that local prompting substantially improves few-shot OOD detection on ImageNet-1k. With four shots, Local-Prompt alone already outperforms CoOp by a large margin, reducing average FPR95 from 33.82 to 27.66 and improving AUROC from 90.98 to 93.53. The gains are consistent across iNaturalist, SUN, and Places, with Texture remaining the hardest OOD dataset for all few-shot methods.

The extensibility of the framework is visible when combining the local branch with LoCoOp. In the four-shot regime, Local-Prompt +LoCoOp improves over LoCoOp from 31.46 to 26.29 FPR95 and

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Table 2: ID accuracy evaluation using ImageNet-1k as ID data.

Method	Tuning Manner	ID accuracy
MCM	*zero-shot*	67.01
GL-MCM		67.01
KNN	*full-tuning*	79.64
NPOS		79.42
LoCoOp	4-*shot*	69.50
Local-Prompt		**70.52**
LoCoOp	16-*shot*	71.86
Local-Prompt		**74.24**

Table 3: Comparison results of near OOD detection tasks. We report 16-shot results for few-shot methods. Our method can achieve performance against other methods.

Method	ID / OOD	ImageNet-10 / ImageNet-20		ImageNet-20 / ImageNet-10		**Average**	
		FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM		5.00	98.71	12.51	97.70	8.75	98.21
GL-MCM		10.10	98.04	9.00	98.62	9.55	98.33
LoCoOp		11.20	97.49	12.00	97.79	11.60	97.64
Local-Prompt		**3.90**	**99.06**	**6.20**	**98.84**	**5.05**	**98.95**

Table 4: Experiments on ImageNet-100. 4-shot results are reported for few-shot tuning.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM	18.13	96.77	36.45	94.54	34.52	94.36	41.22	92.25	32.58	94.48
LoCoOp†	44.94	93.31	30.70	94.70	34.31	93.93	55.92	90.01	41.47	92.99
Local-Prompt	**6.99**	**98.05**	**24.34**	**96.03**	**26.04**	**95.49**	**36.32**	**93.67**	**23.42**	**95.81**

from 92.68 to 94.03 AUROC. Notably, this four-shot hybrid also outperforms the 16-shot LoCoOp model in average OOD detection (26.29 versus 28.66 FPR95, 94.03 versus 93.52 AUROC). In the 16-shot regime, the hybrid again performs best, reaching 24.52 FPR95 and 94.48 AUROC. The per-dataset pattern is also informative. The largest improvements appear on iNaturalist and Places, where crop selection can isolate object parts or scene regions that are weakly represented by a single global score. Texture remains harder, suggesting that some OOD cases are driven by distributed appearance cues rather than a small set of salient local regions.

The same local enhancement also improves ID recognition. As reported in Table 2, Local-Prompt raises ImageNet-1k top-1 accuracy from 69.50 to 70.52 in the four-shot setting and from 71.86 to 74.24 in the 16-shot setting relative to LoCoOp. Table 3 further shows that the local branch is particularly effective on near-OOD transfer: on ImageNet-10 versus ImageNet-20 and the reverse direction, Local-Prompt achieves an average FPR95 of 5.05 and AUROC of 98.95, clearly outperforming MCM, GL-MCM, and LoCoOp. This is consistent with our motivation that near-OOD cases often hinge on subtle regional discrepancies rather than gross global mismatch.

4.3 IMAGENET-100

Table 4 confirms that the benefits are not limited to ImageNet-1k. On ImageNet-100 with four-shot tuning, Local-Prompt reaches 23.42 average FPR95 and 95.81 AUROC, improving substantially over MCM (32.58, 94.48) and LoCoOp (41.47, 92.99). The strongest gain appears on iNaturalist, where FPR95 drops from 18.13 for MCM and 44.94 for LoCoOp to 6.99 for Local-Prompt. This suggests that the local prompting strategy remains effective even when the ID label space is smaller.

5 ANALYSIS

5.1 ABLATION OF THE TRAINING OBJECTIVE

Table 5 decomposes the local training loss. Starting from the positive regional term alone, adding the hard-negative loss improves average FPR95 from 30.85 to 29.27 and AUROC from 92.90 to 93.27. Adding the diversity regularizer yields a further improvement to 28.77 FPR95 and 93.48 AUROC. The progression matches the intended design: the positive term discovers label-consistent regions, the negative term teaches explicit rejection, and the diversity term prevents the negative prompt bank from collapsing to redundant directions.

5.2 NEGATIVE AUGMENTATION AND PROMPT BANK SIZE

Table 6 compares different ways to synthesize hard negatives from ID images. Random crop performs best overall, improving on the baseline and clearly outperforming cutout and Gaussian noise. This is sensible: cropping can remove or distort semantically decisive regions while still preserving realistic image statistics, producing harder negatives than pixel-level corruption.

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Table 5: Effectiveness of each component in loss function.

Loss	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline ($\mathcal{L}_{\text{pos}}$)	11.64	97.59	26.81	94.59	34.50	91.59	50.46	87.84	30.85	92.90
$+\mathcal{L}_{\text{neg}}$	9.84	97.88	24.37	94.97	32.84	91.94	50.02	88.32	29.27	93.27
$+\mathcal{L}_{\text{neg}} + \mathcal{L}_{\text{reg}}$	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

Table 6: Influence of different negative augmentation strategies.

	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline	11.18	97.68	23.30	95.25	30.97	92.44	50.44	87.89	28.97	93.31
+cutout	17.47	96.64	26.17	94.73	34.38	91.84	54.84	86.88	33.21	92.52
+gaussian noise	11.32	97.59	25.70	94.39	34.99	91.71	51.06	88.15	30.76	92.96
+random crop	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

Table 7 studies the number of local negative prompts. Performance is relatively stable between 200 and 300 prompts, with 200 giving the lowest FPR95 and 300 the best AUROC. We use the larger bank in the released implementation because it offers slightly better overall ranking quality while preserving the same qualitative behavior. This stability is useful in practice because it indicates that the method does not rely on fine-tuning the prompt-bank size to a narrow optimum. Once the bank is large enough to cover several distinct negative directions, the regional objective appears to use the additional capacity smoothly rather than overfitting to a particular prompt count.

5.3 Comparison with External Negative Resources

NegLabel uses powerful external negative labels and auxiliary outlier information (Jiang et al., 2024). Table 8 shows that Local-Prompt, trained without outlier data, still achieves a slightly better average result: 24.52 versus 25.40 FPR95 and 94.48 versus 94.21 AUROC. The comparison should be read carefully because the supervision assumptions differ, but it supports the claim that hard negatives synthesized from ID images can be surprisingly competitive.

5.4 Qualitative Evidence

Figures 3–6 provide qualitative support for the method. The density map on iNaturalist suggests improved score separation, the score ablation visualizes the benefit of combining global and local evidence, and the prompt visualizations indicate that Local-Prompt attends to regions that are more diagnostic for OOD rejection. The failure-case examples are particularly informative: several SUN images are difficult to reject from a global viewpoint but become separable once local structures are emphasized.

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Table 7: Influence of different number of local negative prompts.

N_{neg}	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
100	9.02	98.00	24.71	94.82	32.77	91.87	48.81	88.50	28.82	93.30
200	9.65	97.90	22.58	95.26	32.03	92.18	49.85	88.00	28.52	93.33
300	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	28.77	93.48

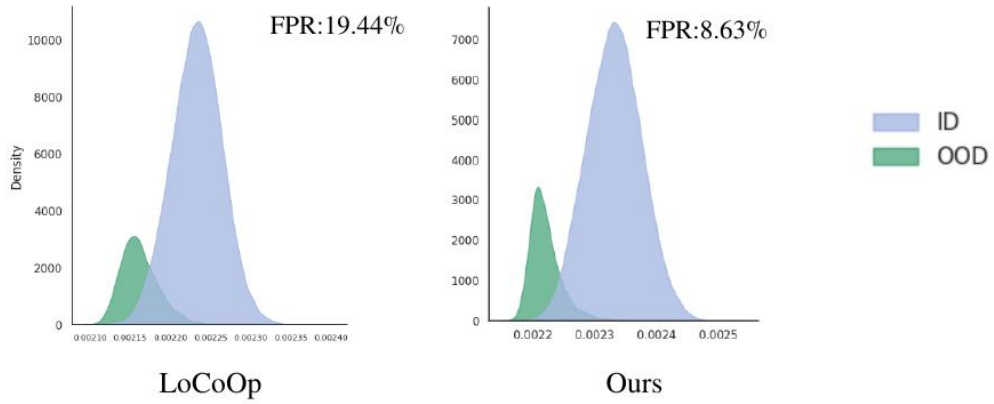

Figure 3: Density map comparison on iNaturalist. The figure visualizes how the proposed score changes the separation between ID and OOD samples relative to global-only scoring.

Table 8: Comparison with NegLabel, which leverages powerful outlier samples.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
NegLabel (w outlier)	1.91	99.49	20.53	95.49	35.59	91.64	43.56	90.22	25.40	94.21
Local-Prompt (w/o outlier)	8.63	98.07	23.23	95.12	31.74	92.42	34.50	92.29	**24.52**	**94.48**

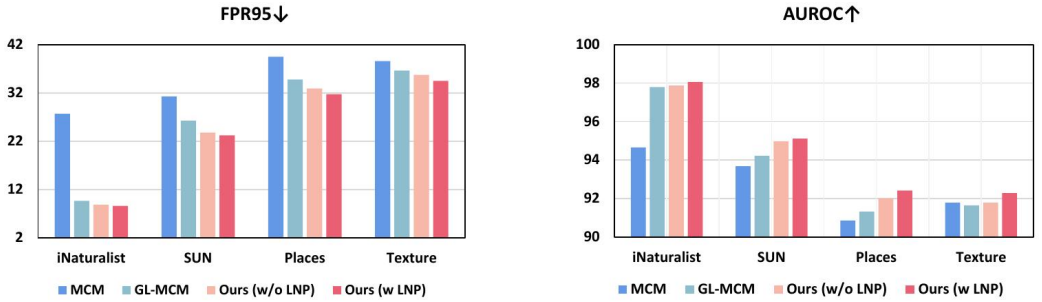

Figure 4: Ablation of different OOD scoring strategies. The visualization complements the quantitative results by comparing several ways of aggregating global and regional evidence.

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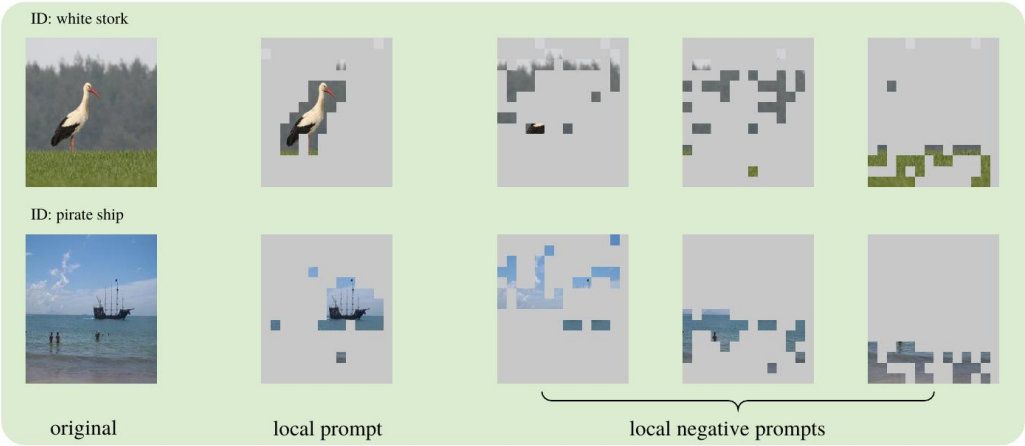

Figure 5: Visualization of ID- and OOD-related regions. The learned local prompts focus on image regions that are more diagnostic for separating in-distribution content from local outlier evidence.

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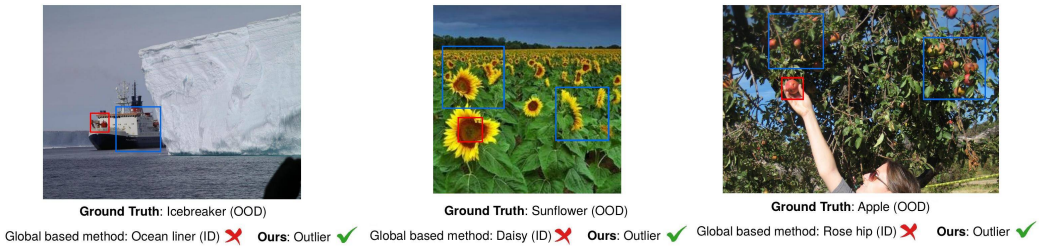

Figure 6: Examples from SUN that are difficult for global-only prompt learning but are correctly separated by Local-Prompt through local enhancement. These cases illustrate the importance of subtle region-level discrepancies.

6 CONCLUSION

We presented Local-Prompt, a few-shot OOD detection framework that shifts prompt learning from the global level to the regional level. The method uses a frozen global prompt branch to mine hard crops, learns class-specific local prompts together with a bank of local negative prompts, and combines both sources of evidence through a regional MCM-style score. Across ImageNet-1k, ImageNet-100, and near-OOD benchmarks, the results show that local prompting consistently improves OOD detection and often improves ID classification as well. The strongest results come from combining the local branch with LoCoOp, which supports the central claim of the paper: regional prompt learning is not only effective on its own, but also an extensible complement to stronger global prompt designs.

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